A Strategy for Structured Illumination in Synthetic Aperture Ultrasound Imaging

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***Abstract*—For many years synthetic aperture ultrasound imaging has been developed for its quality and high-resolution images, but it would be desirable to exceed its resolution limit. Structured illumination by transmitting a series of known grating patterns helps to obtain higher resolution information of the sample in optical microscopy. However, there are many challenges to realize this idea in the ultrasound imaging, regarding the acoustic field manipulation difficulties in synthetic aperture transmissions. This paper presents a method to resolve structures with sub-diffraction scales in ultrasound imaging with the aid of structured illumination. Simulation results show 48 percent resolution improvement compared to the conventional synthetic aperture imaging.**

Keywords—synthetic aperture, super resolution, structured illumination, resolution.

I. Introduction

Structured illumination (SI) is a well-defined method for surpassing the diffraction limit in optical microscopy [1, 2], where imposing grating patterns on the object could help in moving the higher spatial frequencies of the object into the observable spectrum region. For many clinical applications in ultrasound (US) imaging it is desired to resolve structures with sub-diffraction dimensions [3]. Here we employ the SI concept to improve the lateral resolution in synthetic aperture (SA) ultrasound imaging, whereas generating a desired fringe pattern seems impossible in a single element transmission mode such as SA imaging. For this reason, the US transducer transmits randomly apodized plane wave (PW) illuminations for a number of times and receives the backscattered echoes. The recorded backscattered echoes are used to recover the full channel SA dataset. Simultaneously, in order to illuminate the field in a structured manner and achieve higher spatial frequencies, the transmit elements waveforms (wf_{fringe}) should be designed in such a way to collectively generate a PW with the desired periodic fringe pattern, while all elements are active. As a result, in our proposed method, waveform of each transmit element is set as the product of (wf_{fringe}) and that randomly apodized values for each illumination. Compressed sensing theory could help to decrease the number of illuminations without sacrificing resolution. The method can be implemented with conventional ultrasound systems, without the need for additional components.

In the following paragraphs, first we introduce the method for SA data recovery from randomly apodized PW illuminations. After that, acoustic SI for super-resolution PW imaging is described briefly. Based on these two concepts, our proposed method named structured illumination synthetic aperture (SISA) imaging is introduced and the results are compared with PW, SA, and structured illumination plane wave (SIPW) imaging methods.

II. SA Data Recovery from PW Firings

In SA imaging, an individual element is active in each firing and all the elements receive the backscattered echoes, and this process is repeated by choosing each element as the transmitter sequentially [4]. For a PW firing, all the elements are active in each transmission and reception [5]. SA data recovery is realized by transmitting several PWs with different random transmit apodizations. The backscattered echo of each PW firing is a linear combination of the SA dataset [6] and the linear combination coefficients are the transmit apodizations applied in the PW firings. As a result, the SA dataset can be recovered from fewer randomly apodized PW firings with the compressed sensing (CS) reconstruction algorithm [7, 8]. The random apodizations applied in the PW firings compose the measurement matrix of the CS theory. If the number of firings is chosen to be equal to the number of transmitting elements, generally SA data recovery could be accomplished without CS. Final image is reconstructed with the SA dataset.

III. Structured Illumination plane-wave Imaging

Ultrasound lateral spatial resolution is limited to length scales set by the Rayleigh criterion according to:

$$FWHM = const * \lambda \frac{z}{d}, \tag{1}$$

where FWHM is the full width at half maximum of the beam width or point spread function (PSF), λ is the wavelength corresponding to the center frequency of the transmitted beam, z is the distance between the transducer and the focal depth, D is the transducer aperture size and const is a parameter that is determined by the transducer apodization. For a rectangular aperture, const = 1.2063 [9]. Fourier transform of the PSF specifies the lateral spectral bandwidth of the imaging system in k-space, which is limited by cutoff frequency ($\pm k_c$) and has an inverse relationship with spatial resolution. The imaged object has a lateral spectral bandwidth between $-k_o$ and $+k_o$.

Beam formation by an ultrasound transducer introduces a low-pass filter for lateral spatial frequencies thus frequencies above the cutoff frequency cannot be observed. This cutoff frequency limits the lateral resolution. In the SIPW method, to retrieve the high frequency data, a periodic pattern is transmitted to the object. In the spatial domain with a finite aperture a laterally windowed cosine pattern is transmitted, resulting in three delta (narrow Sinc) functions in k-space. When we superimpose a periodic pattern at the object's position, the resulting scattered wave from the object is the product of the insonation pattern and the scattering function of the object. In k-space, this multiplication becomes a convolution resulting in a mixing of the frequencies of the original data and the pattern. As a result, two additional duplications of the k-space information will be generated, each offset to be centered on $\pm k_1$ (where k_1 is the frequency of generated fringe pattern). The echoes undergo a rectangular low-pass filter in k-space with a cutoff frequency of k_c during the receive beam formation process. The received echoes also contain high-frequency components of the original object that have been downshifted due to the fringe pattern illumination. The super-resolution image is reconstructed using computer post processing, which restores the downshifted frequencies by multiplying each received image by an idealized pattern with the same k-space frequencies of 0, $\pm k_1$ [10].

In k-space, after this multiplication, two additional duplications are generated. When capturing only one image, the result is composed of the sum of the three contributions, and thus artifacts that stem from undesired duplications are included in the result. Therefore, a perfect super-resolution image cannot be assembled from a single image, as the frequency components are not separable. To accurately reconstruct the object, a set of images are acquired where the pattern is shifted laterally in spatial domain between subsequent pulses. The shifts are chosen such that the scan covers a full period of the pattern. Typically, three shifts are used to maximize acquisition speed, or five shifts are chosen to increase super resolution image quality. SIPW method is capable of reconstructing objects accurately up to $k_o < k_c + k_1$. In addition, the transmitted pattern itself is limited by the cutoff frequency of the system, and therefore $k_1 \leq k_c$. For the case, where $k_1 = k_c$, the band width of the SIPW imaging transfer function is twice the standard PW imaging.

In summary, SIPW method is as follows [10]: determine the depth to be imaged; determine the shape of the desired pattern containing spatial frequency of the grating pattern and size of the pattern; calculate the transducer elements waveforms (or phases and apodizations) to produce the desired periodic pattern for five shifts to cover a full lateral period; simulate the emitted field to be used in the post processing; excite the transducer with the waveforms; multiply each of the five reconstructed images by the corresponding simulated field; sum and average decoded images; and finally subtract the average of the five SIPW images (without the decoding process) as a low-resolution image from the averaged decoded image to achieve the final high-resolution SIPW image.

IV. Structured Illumination Synthetic Aperture Imaging

Practically, spatial frequency of the periodic pattern cannot exceed cutoff frequency of the imaging system. Above this amount, the degradation in the fringe pattern is noticeable [11], which in turn, distorts the resultant reconstructed image. Therefore, SIPW cannot exceed resolution limit of SA imaging which is theoretically about twice of the PW imaging. On the other hand, structured pattern illumination in SA seems to be inaccessible because fringe pattern generation needs a large

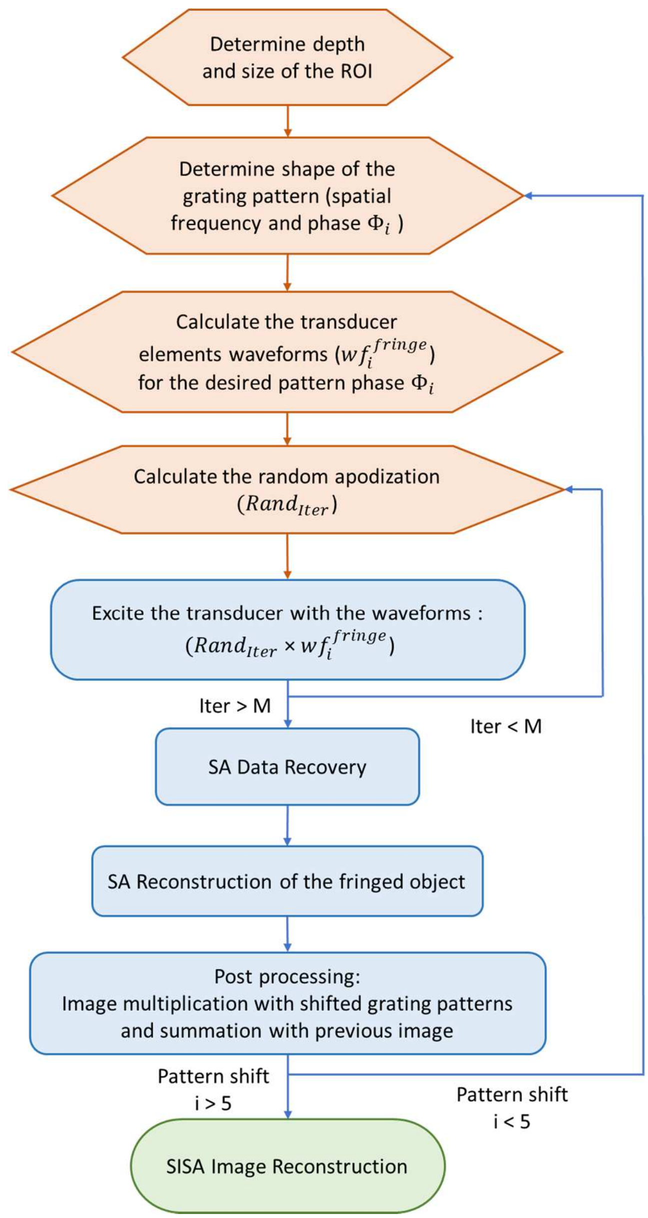

Figure 1: Flowchart of the proposed SISA method. M is the number of random PW firings for SA dataset recovery (M ≤ number of transducer elements). For SI, five pattern shifts are used. $Rand_{Iter}$ contains uniform distributed random variable in the [0,1] interval with the mean value 0.5 ((~U (0,1)).

number of array elements in transmission to make variation in the generated pattern with their different phase and apodizations. The solution is to use pseudo-PW illuminations then recover SA data from PW firings.

In the proposed SISA US imaging method, a fringed object (object multiplied by the grating pattern) image is reconstructed with SA method instead of PW, then post-processing algorithm similar to SIPW is performed on the fringed object image to achieve the high-resolution SISA image (see flowchart of the method in Figure 1). To this purpose, transducer apodization is set as a mixture (multiplication) of the waveforms (or phase and apodizations) for grating pattern generation used in SIPW and random apodized transmissions for SA data recovery. Firings are repeated for M uniform distributed random (~ U (0,1)) apodizations for a specific grating pattern. M could be equal to or less than the number of transducer elements. After M firings, SA dataset is acquired and fringed object is reconstructed with SA method, which results in an object multiplied by the grating pattern. This process is repeated with five phase shifting patterns (Φ_i) and each time post-processing simulated grating pattern with phase Φ_i is multiplied with the relevant result. Finally, all five images are summed to form the SISA image. Unlike SIPW, in SISA there is no need to subtract the low-resolution image from the decoded image.

V. Simulation Results

A. Fringe pattern generation

To achieve higher spatial frequencies, it is required to generate a grating pattern at the ROI. By controlling phase and amplitude of the transducer elements we can manipulate the transmitted acoustic field. In ultrasound, various methods like pseudo-inverse method [12] and Gerchberg–Saxton phase retrieval algorithm [13] have been suggested for pattern shaping in other applications. Multi line transmission (MLT) [14] is a pulse echo imaging method that generates multiple foci by coherently summing the phases of individual foci. Among these methods, MLT is more accurate and could be better performed in pulse echo imaging.

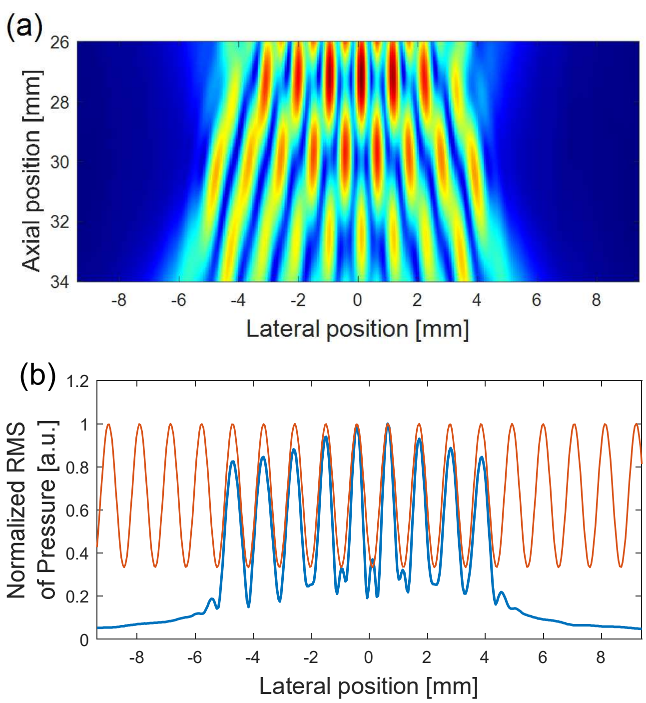

Figure 2: Simulated acoustic field of phased-array transducer (P6-3) for the grating fringe pattern with the period of $\Delta d = 1.1\ mm$ ($k_x = 0.9\ cy.mm^{-1}$). (a) 2D acoustic field pattern. (b) Lateral simulated MLT pattern overlaid by an ideal Sine function with similar frequency.

In Figure 2 acoustic field is simulated by field II software [15, 16] for the MLT method. A phased array transducer (P6-3) was used with 128 elements and pitch size of 0.22 mm. Center frequency was set to 3.6 MHz and sampling frequency was four times of the center frequency. Transmit pulse bandwidth was 60 % (-6 dB) and pulse length was 1 cycle. The speed of sound was set to 1540 m/s. For pattern generation with MLT method, transducer waveforms are designed in such a way to focus at nine points laterally located with 1.1 mm spacing from each other. Lateral spatial frequency (k_x) was set 40 percents of the spatial frequency ($k_x = 0.4\ k = 0.9\ cy.mm^{-1}$) according to the geometrical properties like focus point and aperture size.

B. Single point target imaging

For in silico evaluation, a single point was simulated and imaged with PW, SIPW, SA, and SISA methods in the depth of 30 mm. For SIPW and SISA imaging, the grating pattern spatial frequency is 0.9 $cy.mm^{-1}$ with five foci points according to the size of ROI. Lateral FWHM values are compared in Figure *3* (e). SIPW and SA methods have FWHM values almost close to each other, due to the similar band width of their lateral transfer functions. Among these methods, SISA shows the best FWHM value. Except for the PW imaging, rest of the methods do not have a sidelobe level higher than 0.1 (-20 dB) of the main lobes. Although, side lobe level of the SA method has the lowest value, and SISA's sidelobe is equal to the SIPW but closer to the main lobe.

VI. Conclusion

In US imaging applications, it would be desirable to resolve objects with sub-diffraction scales. Acoustic SI is a candidate to help us reach above the resolution limit in US imaging. Previously reported acoustic SI imaging in the PW framework showed resolution improvement rather than PW imaging. SISA shows that it is possible to exceed the resolution limit of SA US imaging. According Table 1, SISA shows 122 % resolution improvement compared to the PW imaging. These values are for grating pattern with period of 1.1 mm and could be different for other grating pattern periods. Maximum spatial frequency of the grating pattern is limited by the lateral cutoff frequency of the transducer in transmission mode.

One drawback of the proposed method is the large number of required illuminations which limits the frame rate. Although higher number of illuminations will result in better signal to noise ratio (SNR) and contrast to noise ratio (CNR), SA dataset can be recovered from fewer PW firings with the aid of CS reconstruction algorithm.

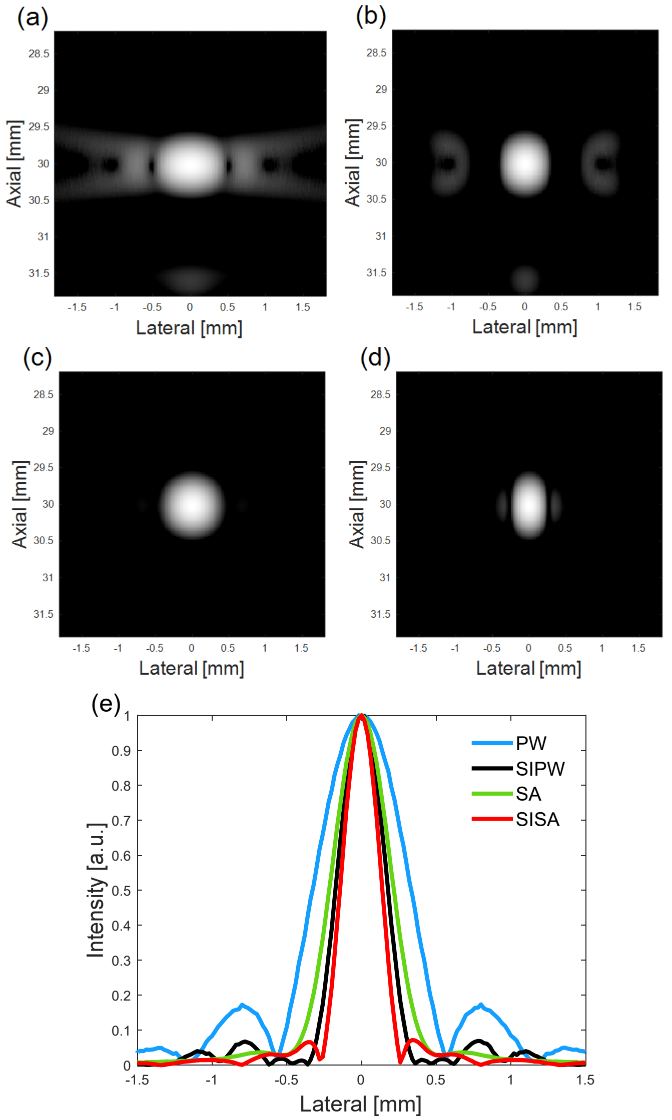

Figure 3: Single point imaging results with (a) PW, (b) SIPW, (c) SA, (d) SISA methods. Dynamic range is 50 dB. Grating pattern period used in SIPW and SISA is $\Delta d = 1.1\ mm$, (e) lateral FWHM values of the four methods.

Table 1:FWHM value comparison of the methods. Grating pattern spatial frequency used for the SIPW and SISA is $0.9\ \mathrm{cy.mm^{-1}}$.

Method	PW	SIPW	SA	SISA
FWHM (mm)	0.631	0.365	0.418	0.282
FWHM Improvement compared to PW	--	73 %	51 %	122 %

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